I grew up in a hospital compound where my father worked. I volunteered there and saw early what poor data infrastructure looks like from the inside: paper forms, inconsistent terminology, information that never made it to the people who needed it. That background made me interested in healthcare systems long before I had any technical tools to address them. The more formal version of that interest started when I wrote a piece during my bachelor's on how AI models trained on clinical data from predominantly white male trial subjects would inherit and propagate those biases at scale. Health inequality is not new, but the idea that AI systems would systematise it further if the field did not take it seriously seemed worth saying. That was the beginning of taking healthcare AI as a direction, not just an interest.

Coming to Uppsala for my master's, I tried to build on that deliberately. I took a course on designing health innovations, which covered the regulatory and business realities of bringing a digital health product to market under EU MDR, and where I served as compliance officer for a group developing a perinatal mental health application. Separately, a course structured as a real consulting engagement placed our cohort inside a project with a healthtech startup working on EHR interoperability at Akademiska Sjukhuset. I spent most of that project on the human research side, conducting interviews with clinicians and IT staff about where the gap between the vision of interoperability and clinical reality actually sits. Both courses were choices. I wanted to understand what the healthcare AI ecosystem in Sweden looks like from multiple angles before settling on where to go next.

The most direct signal of where I am heading is a small feasibility pilot I started this month with the Haematology and Oncology department at King Faisal Specialist Hospital in Riyadh. It is very early. The focus is on whether an AI layer can assist in interpreting capillary electrophoresis outputs for protein analysis, reducing the turnaround time, which currently runs five to seven days due to manual review. Working through the clinical literature to scope it, I learned that several hospitals have moved from capillary electrophoresis to mass spectrometry for haematological protein work, and that AI integration for simpler reads is already being explored in that space. The instruments differ, but the underlying problem structure is similar enough that the domain thinking transfers somewhat. On the technical side, my master's thesis has given me solid experience with transformer architectures and probabilistic classification, though the domain there is content moderation, not proteomics. However, the thesis topic was one I advocated for, knowing I wouldn’t have domain support from my supervisor. This meant learning about a new field and reaching out to researchers and government bodies to better understand it, under minimal supervision.

I have not studied mass spectrometry or omics data formally. That is the real gap. I offer a genuine and long-standing interest in the problem space this research addresses, some early applied work in adjacent clinical AI, and a technical foundation in deep learning for sequence and classification tasks. After this, I want to work in applied AI for healthcare R&D, in whatever form that takes. A PhD developing probabilistic deep learning for mass spectrometry interpretation, within a programme as well-funded and internationally connected as DDLS, is a direct path toward that.